LEVEL 5

COMPUTING PROJECT

Student Guide

Modification History

Version	Revision Description
V1.0	For issue
V2.0	Inclusion of Reflective Practices (September 2019)
V2.1	Updated NOS
V3.0	Revised July 2023
V4.0	Major Refresh

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1. Unit Overview and Objectives

The aim of the unit is to provide a student with the opportunity to employ the skills necessary to develop a computing artefact in the context of a loosely specified problem, which involves research, analysis, design, coding, testing and project management knowledge and skills.

Whilst most projects will be software development of one form or another, it is possible that students may wish to apply other ***computing skills and knowledge*** to solve problems that do not follow the SDLC. For example in the areas of Data Science, AI or computer networking. Such projects can meet the learning outcomes and assessment criteria, and consequently should not be ruled out. However, it is likely that the problem solving methods, models developed and testing processes will be different from those outlined in the following materials. In these cases, the tutor should suggest any modifications required and agree these with NCC Education before the project is started.

Through the combination of project sessions, private study and tutorial supervision on this unit, students will produce for assessment a proposal, report, demonstration and presentation of their project.

2. Learning Outcomes and Assessment Criteria

Learning Outcomes; The Learner will:	**Assessment Criteria;** The Learner can:
1. Identify a suitable computing artefact and development method	1.1 Discuss project rationale, aims and objectives. 1.2 Select and justify an appropriate computing artefact to develop.
2. Project manage the analysis, design, development and, where applicable, deployment of a computing artefact.	2.1 Select and justify the use of an appropriate development method 2.2 Produce a project plan 2.3 Monitor and evaluate progress against a project plan 2.4 Identify and prioritise project risks.
3. Carry out the analysis for a computing artefact	3.1 Elicit and prioritize requirements 3.2 Produce a requirements specification 3.3 Produce appropriate analysis models.
4. Design and develop a computing artefact	4.1 Produce a design specification 4.2 Select and use appropriate tools to construct a computing artefact
5. Test a computing artefact	5.1 Develop appropriate test plan. 5.2 Test that a computing artefact meets its requirements

6. Systematically evaluate the activities undertaken to fulfil the aim of the project.	6.1 Identify success criteria and judge strengths and weaknesses of the project process and outcomes. 6.2 Reflect on the process of conducting the project, identifying learning achieved. 6.3 Articulate and demonstrate the outcomes of the project.

3. Syllabus

Syllabus			
Topic No	Title	Proportion	Content
1	Introduction	1/12 2 hours project session 2 hours of tutorials	• Choosing your project • Appropriate artefacts • Planning your project • Development methods • Risk management • Your Project Proposal ***Learning Outcomes: 1 & 2***
2	Analysis Specifications	1/12 2 hours project session 2 hours of tutorials	• SDLC Review • Requirements Elicitation • Requirements Specification • Analysis models • Use Cases • System architecture • Documentation for YOUR project ***Learning Outcome: 3***
3	Design Specifications	1/12 2 hours project session 2 hours of tutorials	• Design and Development ○ Computational Thinking ○ Design models ○ User Interface Design ○ Design Patterns and Frameworks ○ Selecting appropriate tools and environment. • Design & Development Documentation ○ Structure ○ Contents ***Learning Outcomes: 4***

4	Test Scripts	1/12 2 hours project session 2 hours of tutorials	• Types of testing • Appropriate types of testing • Black box / White box/ Grey box • Unit tests/ Integration tests/ System tests/ Acceptance Tests • Test plan • Test scripts • Documenting tests ***Learning Outcome: 5***
5	Planning the Final Report	1/12 2 hours project session 2 hours of tutorials	• Structure of Final Report • Content of Final Report • Citations and Referencing (Reminder) • Appropriate Appendices • Presentation and Demonstration requirements. ***Learning Outcomes: 1 -6***
6 -12	Project and Report Completion	7/12 116 hours Private study 4 hours (minimum) meetings with tutor	• Private study time should include weekly meetings with your tutor to discuss your progress. • Project production • Project presentation and demonstration. ***Learning Outcomes: 1 - 6***

4. Related National Occupational Standards

The UK National Occupational Standards describe the skills that professionals are expected to demonstrate in their jobs in order to carry them out effectively. They are developed by employers and this information can be helpful in explaining the practical skills that students have covered in this unit.

Related National Occupational Standards (NOS)
Sector Subject Area: ICT Practitioners **Related NOS:** ESKITP4024, ESKITP4034, ESKITP4054, ESKITP4055, ESKITP4064

5. Resources

Lecturer Guide: This guide contains notes for lecturers on the organisation of each topic, and suggested use of the resources. It also contains guidance on how to run the activities presented in the Student Guide.

PowerPoint Slides: These are presented for each topic for use in the lectures, which form part of the Project Sessions. They contain a brief overview of the key concepts. Handout versions of these slides are also available for distribution to students.

Student Guide: This guides the student through the project and presents details of the tasks they should undertake during private study time in order to complete their project.

Project supervisors should direct students to additional relevant resources (books, journal articles, websites etc.) according to their particular projects.

6. Pedagogic Approach

Suggested Learning Hours					
Lecture:	Tutorial:	Seminar:	Tutor Supervision:	Private Study:	Total:
5	10	5	4	176	200

The teacher-led time for this unit is comprised of lectures and seminars ('project sessions'), with private study and tutorial supervision allowing students to complete their projects as required.

6.1 Project Sessions

Project sessions are a mixture of lectures and seminars. There are five project sessions, each of two hours in duration. Whilst these occupy the first five topics for the unit, they need not be scheduled in consecutive weeks, it is at the discretion of the tutor to use them appropriately. It is expected that students would apply the learning to their project at the point of delivery.

Every project session will contain a lecture coupled with either a seminar-based discussion or an exercise. The PowerPoint slides are presented for use during this time. A key point to note is that the lectures are not designed to present new content to students beyond the structure and content of the project report, and an overview of the required depth of the project content. All other content is covered in the other units on the Level 5 Diploma in Computing programme (and specialisations) and is included here to highlight its relevance to the completion of a computing project.

6.2 Supervision

Students should have access to personal tutorial supervision every week for the full duration of the project (i.e. both during and beyond the taught element of the unit). During this time they should meet with their supervisor to discuss their projects and seek support and guidance. Students should meet with their tutor individually for a **minimum** of 4 hours over the course of the project, it is expected that many students will require more supervision time than this.

The role of the project supervisor is to:

- Advise the student about the scope and viability of the work outlined in their project proposal;

- Direct the student to relevant books, papers and other sources of information that will support them in the development of their project;
- Advise the student on the Harvard style of referencing and the penalties associated with plagiarism;
- Maintain regular supervisory contact with the student throughout the whole duration of the computing project;
- Assist the student in managing the timetable of the project and ensure students set SMART objectives for each period;
- Assist the student in identifying when problems are liable to be encountered and how they might be tackled;
- Ensure the student is made aware of inadequate progress or standards of work below the expected level;
- Advise the student on the relevant structure of their project report, demonstration and presentation if it differs from that described here.
- Read and comment on draft sections of the project report, where this is requested by the student, and return such work with constructive criticism and in reasonable time. However, they will **not** proof read all your Final Project Report the week before it is due!

6.3 Private Study

In addition to the taught portion of the unit, students will also be expected to undertake private study. This will consist of carrying out particular activities associated with their project and writing up various sections of the project within appropriate timescales. Details of the activities are provided in the Project Guidelines (see Appendix 1), which are also reproduced in the Student Guide.

7. Assessment

This unit will be assessed by means of a project based assignment worth 100% of the total mark. It is broken down into 4 tasks: The Project Proposal, The Final Report, The Demonstration and Presentation. The assessment will cover the learning outcomes and assessment criteria given above. For a complete assessment breakdown, see *Appendix 2*.

8. British Computer Society Project Criteria

The BCS expect Computing students to undertake a major computing project, as an individual activity, giving them the opportunity to demonstrate:

- their ability to apply practical and analytical skills present in the programme as a whole;
- innovation and/or creativity;
- synthesis of information, ideas and practices to provide a quality solution together with an evaluation of that solution;
- that their project meets a real need in a wider context;
- the ability to self-manage a significant piece of work;
- critical self-evaluation of the process.

Whilst not mandatory requirements, these criteria represent good practice for computing projects.

8.1 Project reports

BCS require that projects must involve the production of a report which should include:

- elucidation of the problem and the objectives of the project
- an in-depth investigation of the context and literature, and where appropriate, other similar products
- where appropriate, a clear description of the stages of the life cycle undertaken
- where appropriate, a description of how verification and validation were applied at these stages
- where appropriate, a description of the use of tools to support the development process
- a critical appraisal of the project, indicating the rationale for any design/implementation decisions, lessons learnt during the course of the project, and evaluation (with hindsight) of the project outcome and the process of its production (including a review of the plan and any deviations from it)
- a description of any research hypothesis
- in the event that the individual work is part of a group enterprise, a clear indication of the part played by the author in achieving the goals of the project and its effectiveness
- references

9. Further Reading List

There is no essential textbook for this unit. However, a selection of sources of further reading around the content of this unit must be available in your Accredited Partner Centre's library. The following list provides suggestions of some suitable sources:

Dawson, C. (2015). *Projects in Computing and Information Systems: A Student's Guide: A Student's Guide (3rd Edition).* Pearson Addison Wesley.
ISBN-10: 1292073462
ISBN-13: 978-1292073460

Weaver, P. (2010). *Success in Your Project: A Guide to Student System Development Projects*. Pearson Education. (ebook)
ISBN-10: 1280600918
ISBN-13: 978-1280600913

Topic 1: Introduction

1.1 Learning Objectives

On completion of this topic, students will be able to:

- Identify a suitable computing artefact and development method.
- Project manage the analysis, design, development and deployment of a computing aretfact.

1.2 Pedagogic Approach

This project session consists of a lecture and a seminar exercise. The lecture provides an overview of development methods and risk management approriate to the computing project.

1.3 Timings

Project Session:	2 hours
Tutorial:	2 hours
Private Study:	12 hours

1.4 Project Session

The Topic 1 Project Session includes a lecture and a seminar exercise.

1.4.1 Lecture Notes

The structure of this topic is as follows:

- Choosing your project
- Appropriate artefacts
- Planning your project
- Development methods
- Risk management
- Your Project Proposal

1.4.2 Seminar Exercises

Task 1

This question is related to the use of Gantt charts and project planning.

If you are not familiar with Gantt charts, or need a refresher, the following resources are useful:

- https://www.gantt.com/
- https://www.wrike.com/blog/project-management-basics-beginners-guide-to-gantt-charts/#Tracking-project-success

A small software house is contracted to develop an Android app for a national environmental charity. The app records surveys of wildlife and updates a cloud database. The database is accessible to PC users via a web browser. An outline list of activities has been identified:

	Activity	Weeks
A	Create detailed requirements specification.	4
B	Design database.	2
C	Develop and test cloud database.	3
D	Develop Web interface.	2
E	Develop Android SQLite database and synchronise with cloud.	3
F	Design and develop Android survey interface and user test.	5
G	Develop app functionality.	4
H	Unit test app/website.	3
I	Integration test.	3

Tasks B and F are all dependent on Task A.
Task C is dependent on Task B.
Tasks D and E are dependent on Task C.
Task G is dependent only on Task E.
Task H is dependent on Tasks D, F, and G.
Task I cannot start until Task H is completed.

a) A Gantt chart will be useful in this project. Explain the purpose of a Gantt chart and describe its components.

b) Draw a Gantt chart for the project activities identified above. Make sure you provide a key to indicate your use of shading, arrows and any other annotation.

c) Explain what is meant by the term 'free float' and illustrate your answer with examples from part b).

Task 2

a) Write down outlines of three ideas for projects. For each of them identify potential artefacts and aims/ objectives.

b) Think about each in turn and identify strengths and weaknesses of each project.

c) Discuss these with another student to help explore the ideas further. At the end of this you should have a better idea of which project(s) are most feasible.

d) Discuss the best two ideas with your tutor, to find out if they are of suitable complexity and viable as computing projects. Make sure you know what you need to do next to develop the ideas further towards a project proposal.

Task 3

Read the following **extract** from a project proposal.

INTRODUCTION

This project is a tool used for tracking the Stationery request for a small company named ABC Spares Ltd. At present the requests for stationery are tracked through traditional methods such as email which leads to an excessive waste of resources.

Therefore, to reduce such wastage of resources and provide a high-level view of the stationery request status to the higher management the stationery request tracking tool for ABC Spares Ltd has been developed.

This system will include various functional components such as Admin, User/Employee, Approvers and a hierarchical structure of employees based on their roles. This system will allow employees to login, check the availability of stationery items and request new stationery items. Approvers will have the ability to approve or reject the stationery requests. In addition, this system will also include features such as automatic notification, a help page and a reporting mechanism for the manager.

The system will also include a database containing details of the employees, their roles, stationery items etc.

In conclusion, with the help of this tool, ABC Spares can provide insights to the higher management for the betterment of the organization.

PROBLEM STATEMENT:

The current system of managing stationery requests involves a lot of paperwork and permissions, resulting in inconsistent data. Multiple requests and explanations are often necessary before approval

for requested products is given. This process also leads to data redundancy as a high percentage of data is copied from other sources. Storage of data in files takes up a lot of space, and it can be challenging to locate a particular record, requiring a lot of time and effort. Additionally, data security is at risk since paperwork can be lost, stolen, or destroyed, and there is no backup for the data. Losing data can result in significant losses for the company.

AIM & OBJECTIVES

The aim of the project is to develop a web-based stationary management system that efficiently and accurately tracks stationery requests in ABC Spares Ltd. while reducing wastage of resources and providing valuable insights to higher management. The objectives of the system are to provide a secure tool that only allows authorized staff to access and approve requests, to enhance fast and easy data retrieval and approval processes, and to reduce the organization's reliance on a file system. The system's hierarchy enables employees to log in and request stationery according to their ranks, and the system ensures accuracy and security by allowing only registered people, including approvers who can also be requesters.

a) Using the guidance provided for your project proposal, analyse the extract against the sections: Background and Aims/ Objectives. Answer the following:

    1) How well does it meet the requirements for the *background* section? Is anything missing? How clear is it? Do you understand the problem, and the artefact to be created? Is the scope clear? What are the strengths and weaknesses of this extract?

    2) How clear are the stated Aims and Objectives? Are the obectives SMART? Is anything missing?

    3) Now use the grading criteria for these two sections and give a grades for Aims/Objectives and Background. Write brief reasons why you awarded these grades.

    4) Discuss your assessment of this extract with another student. How closely did you agree with your grades and assessment?

    5) What feedback would you gove to the student to improve?

1.5 Private Study

The time allocation for private study in this topic is expected to be 12 hours.

Task 1

Refer to your project guidelines. You should use this private study time to think about the project you are going to develop and to write a project proposal, using a template or format that is acceptable to your tutor.

Be prepared to discuss your project ideas with your tutor and to show your work in progress. Your tutor will guide you on the scope of your project and whether the proposal is acceptable.

By the end of this topic, you should have completed the following:

- Thought of an initial project idea
- Found or developed a suitable project proposal template
- Discussed your initial project idea with your tutor
- Produced an outline draft Project proposal, containing Background, Aims, Objectives, methods.

Reflective tasks:

Task 2:

What have I achieved with my work on the *Computing artefact project management* topic this week?

Things to consider:

- Have I practiced a new skill?
- Have I learned something new?
- Have I made any connections to other topics/the wider world with my learning?
- As a result of this, I now need to....

Task 3:

Which were the most difficult parts, and why were they difficult for me?

Things to consider:

- Have I had any prior knowledge of this topic?
- Have I conducted any wider reading?
- How much time did I devote to understanding these parts?
- Consequently, I am now required to....

Note: To be more prepared for addressing the above reflective tasks questions, please refer to the 'Reflection' guidance section below. In there, you will be introduced and encouraged to use several theoretical models of reflection (such as Rolfe et. al's Reflective Learning Framework, Gibbs Reflective Model and so on) that can help you examine all aspects of your learning experience at sufficient depth.

1.6 Reflection

Reflection is the active process of reviewing one's own experiences so that they can be investigated further. Research suggests that it is vital to focus reflection on specific aspects to achieve a solid understanding of the various factors which impact learning. Also, research has shown the influence of reflective writing on the development of reflective practice in many fields such as teaching, medicine, nursing and social work. Reflection can explicitly reveal the implicit knowledge underlying the learning practice so that it can be identified, debated and improved. In addition, reflection enables us to draw on the past and present and guide ourselves into a better future. Therefore, some different theoretical reflection models are presented below to help students synthesise and systematically develop their answers to the reflective tasks questions.

To make the best use of the reflective tasks, students should identify the interesting or problematic aspects of learning from the topic. Then, they should investigate those topics and aspects in several ways. For example, students can examine the assumptions they held about the topics before the lesson and how they might have changed after the lesson. Also, students should think about how those assumptions might have affected their receptivity to the new topics, what alternatives there are to overcome any learning issues and what should be done in the future to make learning new topics easier and/or more enjoyable. To get the most out of the critical, reflective tasks, students should be open-minded, responsible and continuously questioning. It may also be beneficial for students to position their learning in a wider context and investigate how their beliefs, morals, norms and values my influence their learning. Ultimately, students should learn more about themselves and their teachers as they engage in a learning experience, which should not only involve acquiring or increasing technical knowledge, but also a preparedness to act for change.

Despite the agreement that there is no one 'right way' to engage in reflective practice, it is agreed that adopting a systematic process which actually leads to building knowledge is important. Therefore, when tackling the reflective tasks questions, students are advised to base their answers around the recommended reflective models.

The below points list some of the prevalent models of reflection and outline the situations they are best used for:

1. Gibbs reflective cycle (1988)

In 1988, Graham Gibbs developed this model to give structure to learning from both standalone and repetitive (cyclical) experiences. This model is fairly simplistic, consists of six distinctive stages and makes the practitioner aware of the stages he/she goes through when experiencing an event. However, a criticism is that these stages might draw shallow reflection on the practitioner's part as there is not a referral to critical thinking, analysis, assumptions or prompt to view experiences from different perspectives.

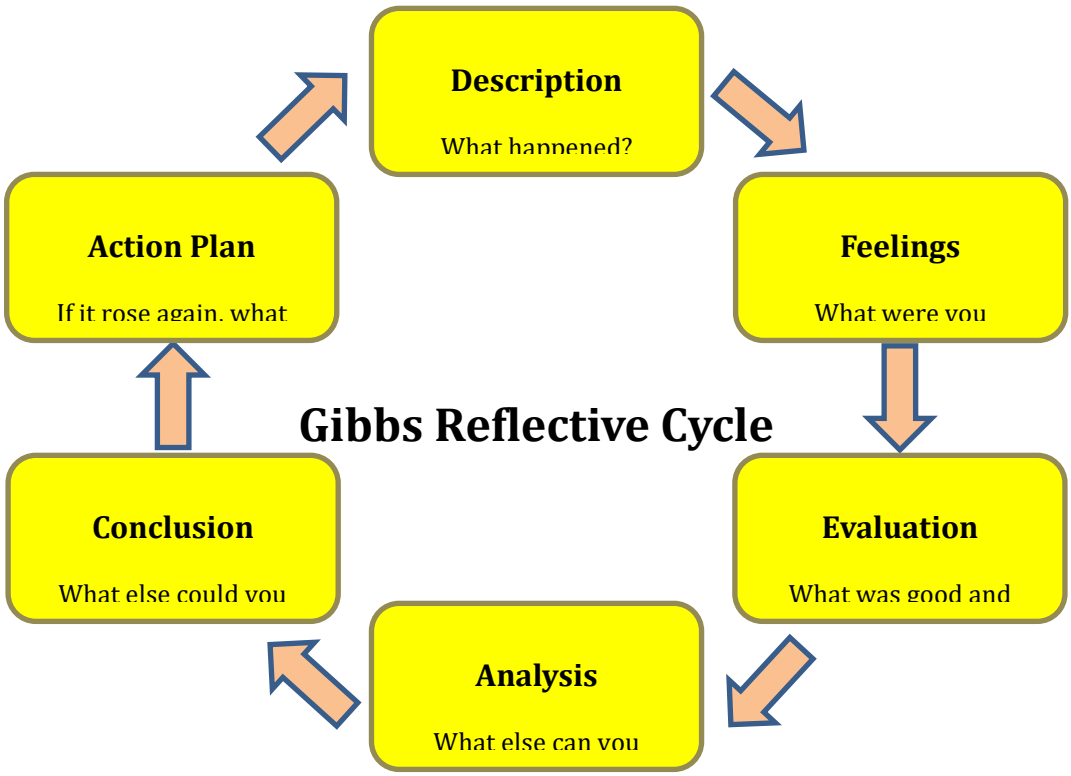


Figure 1: Gibbs reflective cycle

2. Kolb reflective cycle (1984)

In 1984, David Kolb developed the Experiential Learning Theory (ELT) which has ever since been influential in the learner-centred learning. The basis of this model is the practitioner's own experience, which is systematically evaluated and analysed in three stages. This model is good in that it is basic and consists of distinct stages. However, like the Gibbs's model, the stages might draw shallow reflection from the practitioner as they do not refer to critical thinking, analysis, assumptions or prompt him/her to view experiences from different perspectives. Also, it has been criticised for not taking account of the different cultural experiences and conditions, among other criticisms.

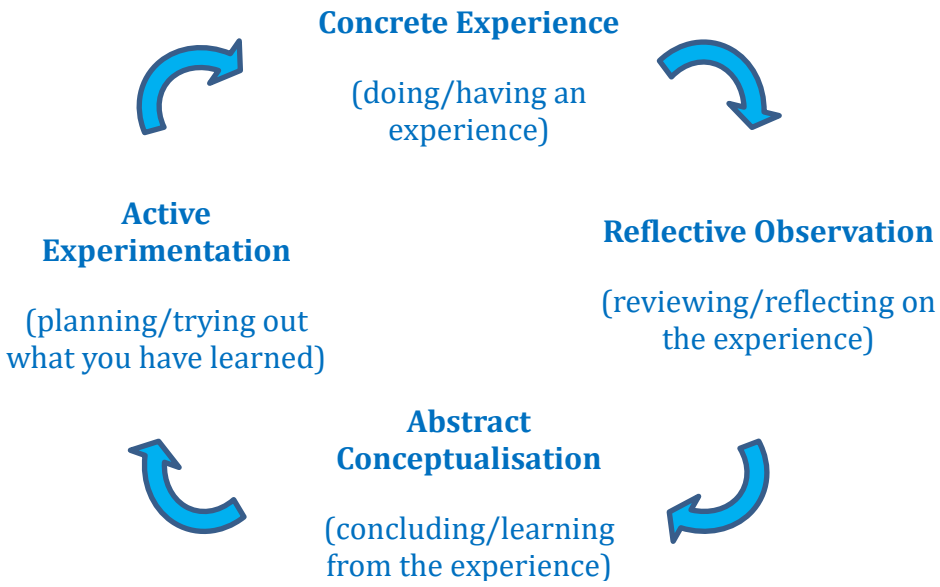


Figure 2: Kolb reflective cycle (1984)

3. Schon Model (1991)

Schön distinguishes between *reflection-on-action* and *reflection-in-action*. *Reflection-in-action* enables the practitioner to identify a problem as it occurs, consider alternatives and alter course of action to solve it, while *reflection-on-action* is the careful consideration of an event after it has occurred. Although this model is professional and highlights the difference between two types of reflection, it does not provide guidance for implementing them.

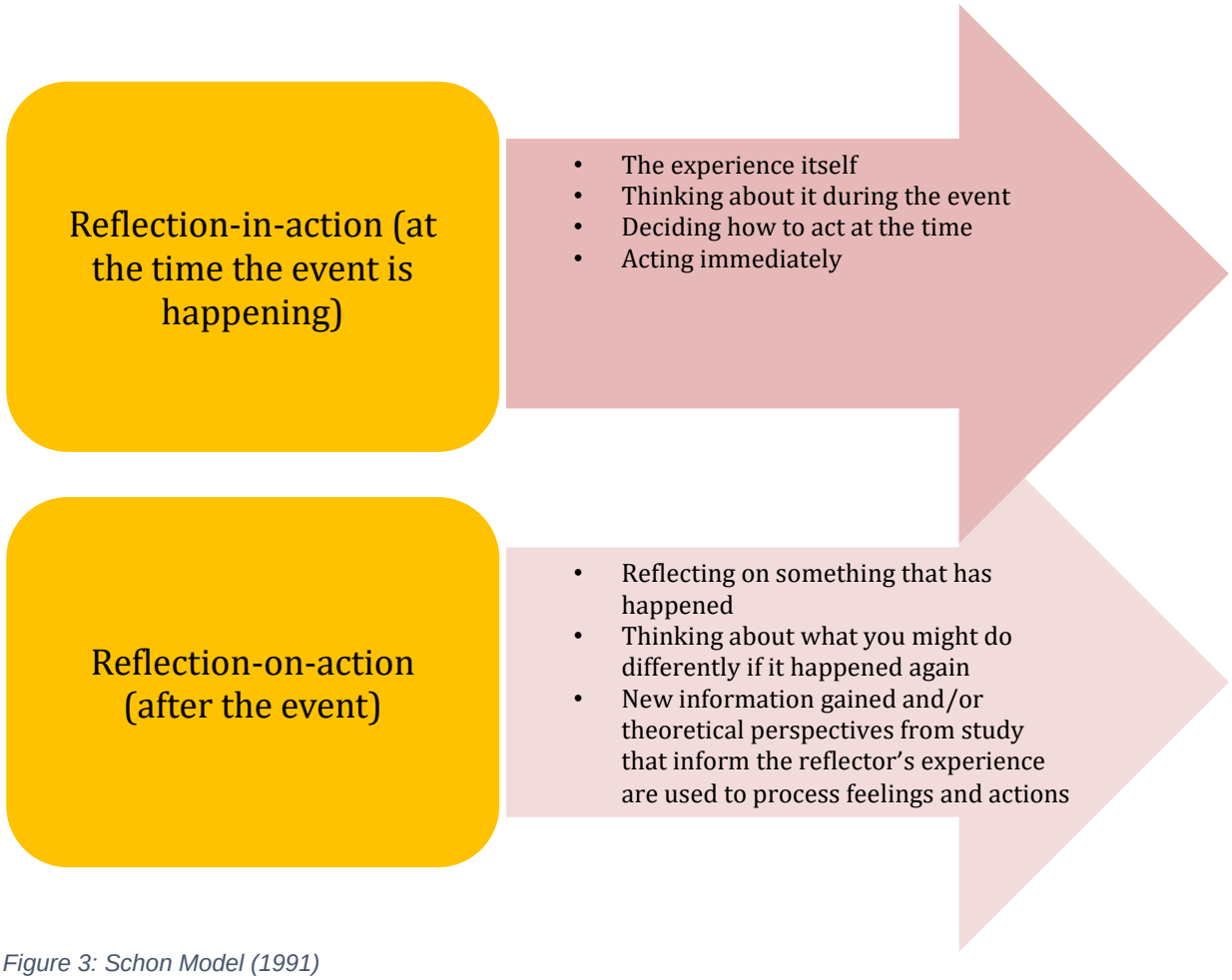


Figure 3: Schon Model (1991)

4. Driscoll model (1994)

This model is based on stages comprising three questions: '***What?', 'So what?' and 'Now what?***' The first stage of the model involves the identification and description of the chosen experience/event. The second stage prompts the practitioner to analyse and evaluate feelings and discuss the effects of actions. The final stage guides the practitioner to apply, synthesise and consider implication and actions. Although the model is an organisational model and is easy to follow and remember, it does not lead the practitioner to deeply reflect about themselves – but only the situation.

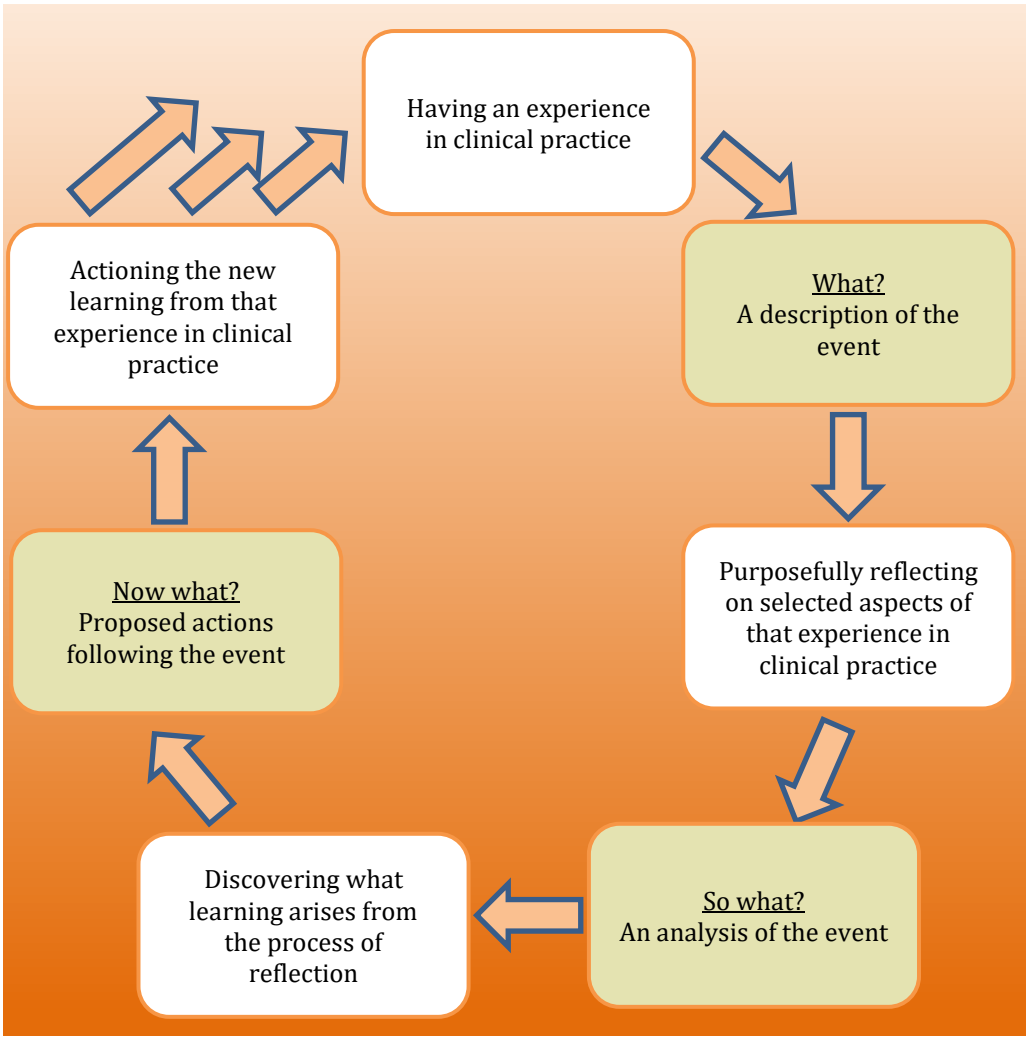


Figure 4: Driscoll model (1994)

5. Rolfe et al. (2001)'s Reflective Learning Framework

This reflective model was originally developed for nursing. However, it is equally appropriate for other situations and is based on three questions: 'What?' (description), 'So What?' (Analysis) and 'Now What?' (Action plan to improve learning/practice). These questions are repeated at three levels with deeper reflection at each level. The levels are also descriptive and action-oriented. This model is good in that by working through the same questions at various levels, the practitioner can advance to expert level. Although the questions need elaboration to direct students to avoid a superficial approach, the model is simple and future focussed.

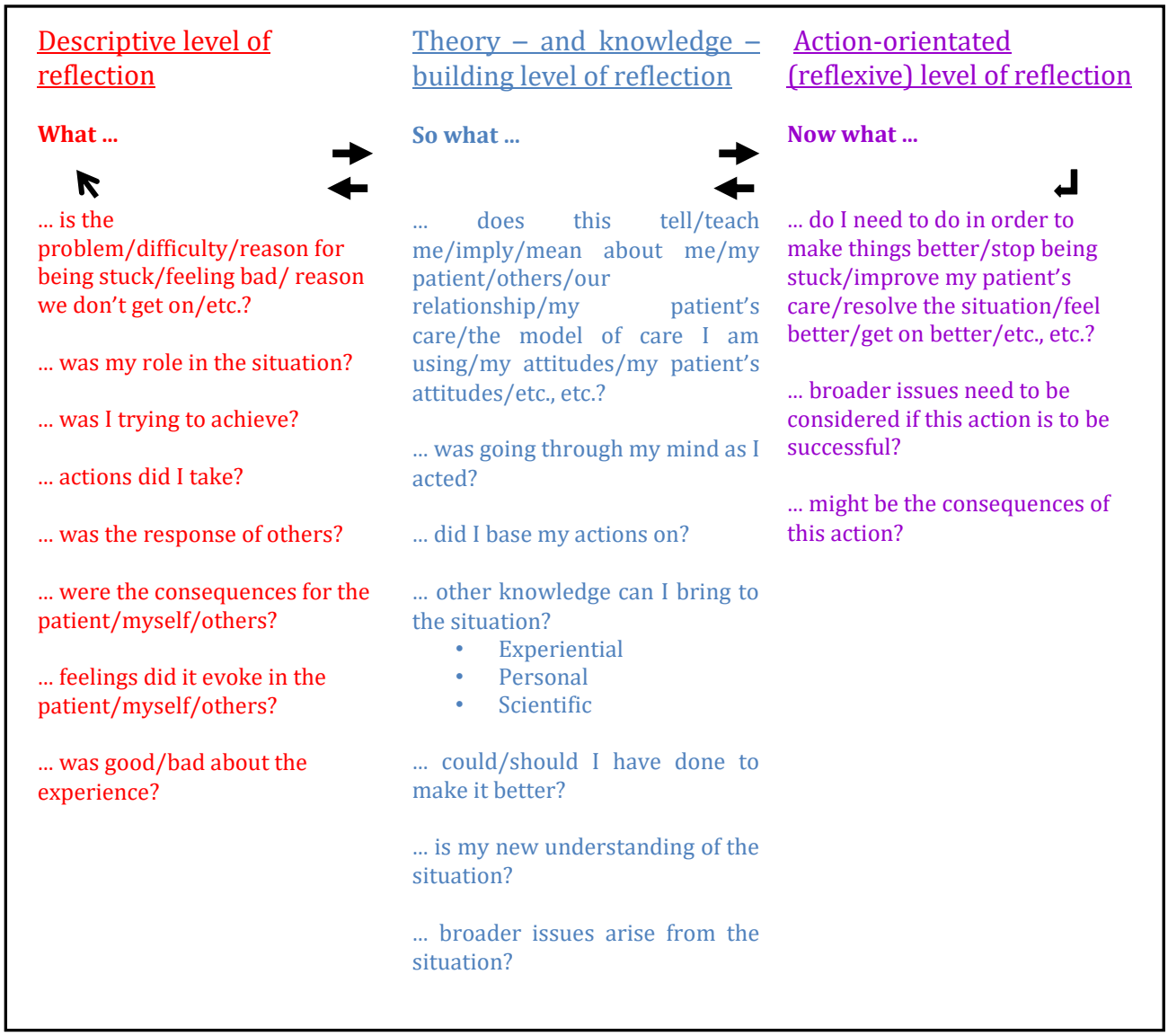


Figure 5: Rolfe et al. (2001)'s Reflective Learning Framework

Reflection: References

Atkins, S. and Murphy, K. (1993) Reflective practice. *Nursing Standard* 9(45): 31–7.

Gibbs, G. (1988) Learning by Doing: A guide to teaching and learning methods. Oxford: Further Education Unit, Oxford Brookes University.

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Kolb, D. A. (1984) *Experiential Learning: Experience as the source of learning and development*. New Jersey: Prentice Hall.

Rolfe, G., Freshwater, D., Jasper, M. (2001) Critical reflection in nursing and the helping professions: a user's guide. Basingstoke: Palgrave Macmillan.

Schon, D. (1983) *The Reflective Practitioner*. London: Temple Smith

Topic 2: Analysis & Requirements Specification

2.1 Learning Objectives

On completion of this topic, students will be able to:

- Carry out the analysis of a computing aretefact.
    - Elicit and prioritize requirements
    - Produce a requirements specification
    - Produce appropriate analysis models suitable for your project

2.2 Pedagogic Approach

This project session consists of a lecture and a seminar exercise. The lecture provides an overview of an anlaysis specification approriate for the computing project:

The seminar exercise provides students with an oppurtunity to discuss the analysis specification and gain a greater understanding of how it relates to their project.

2.3 Timings

Project Session:	2 hours
Tutorial:	2 hours
Private Study:	12 hours

2.4 Project Session

The Topic 2 Project Session includes a lecture and a seminar exercise.

2.4.1 Lecture Notes

The structure of this topic is as follows:

- SDLC Review
- Requirements Elicitation
- Requirements Specification
- Analysis models
- Use Cases
- System architecture
- Documentation for YOUR project

2.4.2 Seminar Exercises

Task 1

Developing your requirement specification and System Use Case diagram

a) Identify who the key stakehgolders are for your project

b) Identify a list of questions you need to ask (information you need to find) in order to produce the detailed functional and non-functgional requirements.

c) Identify where you can find the information (for example which key stakeholder can provide it).

d) Draw a draft use case diagram for the different actors and the main use cases. This should raise some questions about aspects that need clarification. You can add these to your list of questions.

2.5 Private Study

The time allocation for private study in this topic is expected to be 12 hours.

Task 1

Refer to your project guidelines. At this stage, you should have developed a draft project proposal for discussion and review with your tutor. If you have not done this, make sure this is completed and shown to your tutor as soon as possible.

You should also use this private study time for planning to start the analysis of the project based on your draft proposal.

Task 2

Prepare a progress statement to discuss with your tutor during your supervision sesison. This should include:

- The work completed thus far
- A forecast of work to be completed
- An updated project plan

By the end of this topic, you should have done the following:

- Thought of an initial project idea (Topic1)
- Found or developed a suitable project proposal template (Topic1)
- Discussed your initial project idea with your tutor (Topic 1)
- Drafted a project proposal (Topic 1)
- Discussed your draft project proposal with your tutor
- Begun the analysis of your project based on the final project proposal.

Reflective tasks:

Task 3:

What have I got out of doing these activities?

Things to consider:

- Have I developed my understanding from the previous topic?
- Have I managed my time effectively to dedicate myself to the learning?
- Have I learned something that is completely new and if so how will I benefit?
- Therefore, I should now....

Task 4:

How have I developed my knowledge and skills?

Things to consider:

- Was the content of *Analysis Specifications* new to me?
- Has my subject vocabulary increased this week?
- Have I gone beyond just reciting definitions?
- Now that I have reflected, I now need to....

Note: To be more prepared for addressing the above reflective tasks questions, please refer to the 'Reflection' guidance section. In there, you will be introduced and encouraged to use several theoretical models of reflection (such as Rolfe et. al's Reflective Learning Framework, Gibbs Reflective Model and so on) that can help you examine all aspects of your learning experience at sufficient depth.

Topic 3: Design Specifications

3.1 Learning Objectives

- On completion of this topic, students will be able to:
    - Design and develop a computing artefact.
        - Produce a design specification comprising
            - Structural models
            - Behavioural models
            - User Interface model
        - Select and use appropriate tools to construct a computing artefact

3.2 Pedagogic Approach

This project session consists of a lecture and a seminar exercise. The lecture provides an overview of an design and developmentspecification approriate for the computing project.

The seminar exercise provides the students with an oppurtunity to discuss the design specification and gain a greater understanding of how it relates to their project.

3.3 Timings

Project Session:	2 hours
Tutorial:	2 hours
Private Study:	12 hours

3.4 Project Session

The Topic 3 Project Session includes a lecture and a seminar session.

3.4.1 Lecture Notes

The structure of this topic is as follows:

- Design and development :
  - Computational Thinking
  - Design models
  - User Interface Design
  - Design Patterns and Frameworks
  - Selecting appropriate tools and environment.
- Design & Development Documentation
  - Structure
  - Contents

3.4.2 Seminar Exercises

Task 1

a) Take 5 minutes to write down at least two things that you either did not understand from the lecture or that you feel you need further clarification on.

b) Note at least two things that you are very clear on and understand fully from the lecture.You will feed this back to your tutor.

Your tutor will then split you into groups within which you should attempt to resolve all of the things that the group members did not understand.

Task 2

a) Review the material in this lecture and identify which models will be most appropriate for your project. Draft a template for your chapter, with notes of what needs to be included in each section.

b) Explain what is meant by decomposition and refinement, and how you will use that principle in your project.

Task 3

a) Scope creep is often a significant problem in projects. Explain what it means. Identify any aspects of your project where it has ocurred or has the potentiual to occur. Review the scope statement in your report and check if it requires ammendment.

3.5 Private Study

The time allocation for private study in this topic is expected to be 12 hours.

Task 1

Refer to your project guidelines. At this stage, you should have a draft analysis of the system completed.

You should also use this private study time for planning the design phase of the project. Be prepared to discuss the progress of your project with your tutor and to show the written sections which should now be completed.

Task 2

Prepare a progress statement to discuss with your tutor during your supervision sesison. This should include:

- The work completed thus far
- A forecast of work to be completed
- An updated project plan including revised Gantt chart.

By the end of this topic, you should have done the following:

- Thought of an initial project idea (Topic1)
- Found or developed a suitable project proposal template (Topic1)
- Discussed your initial project idea with your tutor (Topic 1)
- Drafted a project proposal (Topic 2)
- Discussed your draft project proposal with your tutor (Topic 2)
- Begun the analysis of your project based on the final project proposal (Topic 2)
- Drafted an analysis of the system you are going to develop
- Discussed the draft analysis with your tutor

Reflective tasks:

Task 3:

What else have I got out of studying the *Design Specifications* topic?

Things to consider:

- Have I worked well with other students?
- Have I sought clarity when required?
- How organised have I been to ensure my learning was successful?
- Having done the reflection task, I now discovered that I need to...

Task 4:

How can my learning be amplified?

Things to consider:

- Can the skills that I have learned be used elsewhere?
- Does this content cross over to other topics/subjects?
- What good practice can I take from my learning and transfer elsewhere?
- This reflection task helped me know that I need to...

Topic 4: Test Scripts

4.1 Learning Objectives

On completion of this topic, students will be able to:

- Develop appropriate tests for a computing artefact
    - Explain the purpose and types of testing
    - Develop appropriate test plan
    - Test that a computing artefact meets its requirements

4.2 Pedagogic Approach

This project session consists of a lecture and a seminar exercise. The lecture provides an overview of testing approriate for the computing project.

The seminar exercise provides the students with an oppurtunity to develop test scripts that can be enhanced to support the testing needs of their project.

4.3 Timings

Project Session:	2 hours
Tutorial:	2 hours
Private Study:	12 hours

4.4 Project Session

The Topic 4 Project Session includes a lecture and a seminar exercise.

4.4.1 Lecture Notes

The structure of this topic is as follows:

- Types of testing (reminder)
- Appropriate types of testing
    - Black box / White box
    - Unit tests/ Integration tests/ System tests/ Acceptance Tests
- Test plan
- Test scripts
- Documenting tests (in YOUR project)

4.4.2 Seminar Exercises

Presented below are two unit test scripts in the format that you should use for the test scripts in your project. Read through the test scripts and then work through the tasks.

Unit Test 1		Tests Class: EmployeeDetails	Designed By: John Smith	
Data Source: User Entry		Objective: Test basic functionality	Tester: John Smith	
Test Case	Description	Tasks	Expected Result	Actual Result
1.1	Test for basic functionality	Enter employee details: EMPLOYEE NUMBER: 123456 FIRST NAME: Steve SECOND NAME: MOSS	Record is added to the database	

Unit Test 2		Tests Class: SalaryDetails	Designed By: John Smith	
Data Source: User Entry		Objective: Test basic functionality	Tester: John Smith	
Test Case	Description	Tasks	Expected Result	Actual Result
1.1	Test for basic functionality	Enter employee details: EMPLOYEE NUMBER: 123456 SALARY: $20,000	Record is added to the database	

1. Convert the unit test scripts into an integration test script.

2. Define a unit test script for the Class JobDetails the inputs are EMPLOYEE NUMBER and JOB TITLE

3. Add the unit test script that you produced in task 2 to the integration test script that you produced in task 1.

4. Using the template below for a SYSTEM test, containing an example of a non-functional requirement (security) create another example.

System Test Suite 1.1		Tests non-functional requirement 2:security of admin login.	Designed By: <Student Name>	
Data Source: User Entry		Objective: Test for SQL injection attack.	Tester: <Student Name>	
Test Case	Description	Tasks	Expected Result	Actual Result
1.1.1	Test for basic functionality	Enter login details: Username: admin Password: unknown’ or'1'='1	Login refused	

5. System tests are often devised at the time the requirement specification is written, since they are testing these requirements. Look at the requirements in your project and draft out some system tests for functional and non-functional requirements.

4.5 Private Study

The time allocation for private study in this topic is expected to be 12 hours.

Task 1

Refer to your project guidelines. At this stage, you should have completed a draft design.

You should also use this private study time for planning how you are going to test your system. Be prepared to discuss the progress of your project with your tutor and to show the written sections which should now be completed.

Task 2

Prepare a progress statement to discuss with your tutor during your supervision sesison. This should include:

- The work completed thus far
- A forecast of work to be completed
- An updated project plan

At the end of this topic you should have done the following:

- Thought of an initial project idea (Topic1)
- Found or developed a suitable project proposal template (Topic1)
- Discussed your initial project idea with your tutor (Topic 1)
- Drafted a project proposal (Topic 2)
- Discussed your draft project proposal with your tutor (Topic 2)
- Begun the analysis of your project based on the final project proposal (Topic 2)
- Drafted an analysis of the system you are going to develop (Topic 3)
- Discussed the draft analysis with your tutor (Topic 3)
- Drafted a design for the system you are going to develop
- Discussed the draft design with your tutor
- Drafted an initial plan for the testing of your system

Reflective tasks:

Task 3:

What was the best thing I did?

Things to consider:

- Have I reviewed any notes that I have made on the *Test Scripts* topic?
- Have I read more widely?
- Have I sought the opinion of others?
- As a result of this, I now need to....

Task 4:

Why was this the best thing I did?

Things to consider:

- Have I consulted wider academic texts which made me more informed about the *Test Scripts* topic?
- Have I consolidated my learning?
- Have I ever done this before?
- Consequently, I am now required to...

Note: To be more prepared for addressing the above reflective tasks questions, please refer to the 'Reflection' guidance section on pages 21-27. In there, you will be introduced and encouraged to use several theoretical models of reflection (such as Rolfe et. al's Reflective Learning Framework, Gibbs Reflective Model and so on) that can help you examine all aspects of your learning experience at sufficient depth.

Topic 5: Final Report and Presentation/Demonstration

5.1 Learning Objectives

On completion of this topic, students will be able to:

- Describe the requirements for their final report
- Complete report sections in a timely manner
- Explain the requirements of the project presentation/ demonstration

5.2 Pedagogic Approach

This project session consists of a lecture and a seminar exercise. The lecture provides an overview of the structure and content of the final report for the computing project.

The seminar exercise provides the students with an oppurtunity to discuss the structure and content of the final report and gain a greater understanding of how it relates to their project.

5.3 Timings

Project Session:	2 hours
Tutorial:	2 hours
Private Study:	12 hours

5.4 Project Session

The Topic 5 Project Session includes a lecture and a seminar discussion.

5.4.1 Lecture Notes

The structure of this topic is as follows:

- Structure of the final report
- Content of the final report
- Citations and referencing (reminder)
- Appropriate appendices
- Presentation and Demonstration requirements

5.4.2 Seminar Exercises

The aim of the seminar session is to provide the students with an opportunity to discuss the structure and content of the final report and gain a greater understanding of how it relates to their project.

This seminar does not have a particular practical exercise associated with it because what is important for this topic is that students understand the structure and content of the project report. The structure of this discussion is left to the lecturer.

One key point to make during the seminar is the need for supportive narrative for every diagram presented in the project report. The point of this seminar is for the lecturer to ensure that the students follow the structure presented in Topic 5 for their project report and stress that marks will be lost if they do not.

This session provides an opportunity to discuss the structure and content of the project report.

5.5 Private Study

The time allocation for private study in this topic is expected to be 12 hours.

Task 1

Refer to your project guidelines. At this stage you should have completed draft unit testing test scripts..

You should also use this private study time for planning what tasks you still need to carry out for your project and to keep writing up your project so that you are on target to meet the hand in deadline.

Be prepared to discuss the progress of your project with your tutor and to show the written sections which should now be completed.

Task 2

Prepare a progress statement to discuss with your tutor during your supervision sesison. This should include:

- The work completed thus far
- A forecast of work to be completed
- An updated project plan

At the end of this topic you should have done the following:

- Thought of an initial project idea (Topic1)
- Found or developed a suitable project proposal template (Topic1)
- Discussed your initial project idea with your tutor (Topic 1)
- Drafted a project proposal (Topic 2)
- Discussed your draft project proposal with your tutor (Topic 2)
- Begun the analysis of your project based on the final project proposal (Topic 2)
- Drafted an analysis of the system you are going to develop (Topic 3)
- Discussed the draft analysis with your tutor (Topic 3)
- Drafted a design for the system you are going to develop (Topic 4)
- Discussed the draft design with your tutor (Topic 4)
- Drafted an initial plan for the testing of your system (Topic 4)
- Discussed the intial testing plan with your tutor
- Finalised the testing plan for the project

At this point ensure that you are up to date and have completed all the tasks above and that everything has been fully documented so that you can easily write up each stage of the project in the final report.

Reflective tasks:

Task 3:

Whilst studying the *Final report* topic, what worked least well for me? Why did this not work well for me?

Things to consider:

- Did I struggle to memorise a concept?
- Did I rely on too few information sources?
- Have I had any experience with *Final report* topic before?
- Therefore, I should now....

Task 4:

What have I learned about the *Final report* topic from this not having worked well for me?

Things to consider:

- Does it require a lot of preparation?
- Do I need to develop a particular academic skill?
- Is it suited to a particular learning style?
- Now that I have reflected, I now need to….

Note: To be more prepared for addressing the above reflective tasks questions, please refer to the 'Reflection' guidance section. In there, you will be introduced and encouraged to use several theoretical models of reflection (such as Rolfe et. al's Reflective Learning Framework, Gibbs Reflective Model and so on) that can help you examine all aspects of your learning experience at sufficient depth.

Topic 6: Topics 6-12: Project and Report Completion

6.1 Private Study

The time allocation for private study is expected to be 120 hours plus time spent meeting with your tutor.

Refer to your project guidelines. You should now complete your project and report, as well as prepare for your project demonstration and presentation.

You should now feel comfortable reflecting on your learning each week, based on the activities you have attempted so far. Please continue to do this as it will stand you in good stead when producing your project.

Throughout this period you are expected to maintain contact with your tutor/supervisor and inform them of the progress you are making and any problems or questions you have. You should continue to prepare progress statements for each meeting with your tutor. You will also be expected to produce evidence of your work and demonstrate continually improving versions of your computing artefact in each meeting with your tutor.

You will need to manage your time carefully to ensure that you continue to work systematically through your project. It is now your responsibility to manage the project plan and ensure you keep making progress each week.

Appendix 1: Project Guidelines

1. Introduction

The Project unit is unlike any of the other units on your diploma. It gives you the opportunity to pursue work on a subject ***you*** are particularly interested in, it gives you a chance to show initiative, use and build on the skills and knowledge that you've developed so far.

This is ***an individual projec***t which must show a good understanding of the subject area and associated tools and techniques. It is a ***Computing project*** and so it.should cover, in detail, the analysis, design coding and testing as outlined in the learning outcomes for the unit. It should also give an overview of the approaches used for project management and risk analysis.

The project gives you the chance to exercise skills that will be important to you in your future career and which are difficult to measure in other ways: to do well you will need to show initiative, motivation and the ability to work in an organised fashion over an extended period. Most students find doing the project a very useful and rewarding experience. In addition, future employers are often very interested in a student's Project.

2. Aims of the Project

The overall aims of the project are that you will be expected to:

- combine previously acquired knowledge and techniques with newly investigated ideas, the results to be communicated in a report;
- Work independently under the guidance of a Supervisor on a project of your choice and be expected to develop it systematically;
- Produce an extended piece of work covering a variety of activities related to a single theme.

3. Choosing a project

The first and often most difficult task is to find a suitable subject for a project. The topic you choose for your project must obviously be one which you find interesting and on which you want to work on over the duration of the unit. A project topic must be of a suitable standard and subject to the approval of the Module leader.

There are several ways of choosing a project topic:

- Develop an idea of your own. If you have an idea that you think would be suitable for a project, go and discuss it with the unit tutor. Even if your initial idea is slightly vague, it is essential that you talk to your tutor - this often has the effect of crystallising your ideas and generating a suitable project topic.
- Do a project suggested by a member of staff. Sometimes tutors have a list of possible projects.
- Do a project suggested by some outside organisation. This is usually the development of a software system for the organisation. Just because you have chosen to do a 'real project', it does not necessarily mean that it is suitable as an undergraduate project, you will have to discuss it with your tutor – and see the additional guidance below about undertaking a project for an external organisation.

Once you have chosen a project topic, you must submit a project proposal (see section below)

4. Organising your Time

As a rough guide the project should take up to 200 hours of your time, in total; this time should be distributed evenly throughout the unit. If you spend much longer than this on your project you will be in danger of interfering with work for other units. Hence any possible benefit gained from producing a slightly better project will be outweighed by the harm done to your marks on other units. If you find that your project is taking up too much of your time, consult your project Supervisor who will be able to advise you.

You are strongly recommended to keep a project diary, in which you record all the work, activities and decisions made during the project. This will help you to keep to your work plan, and also be a useful source of material when you come to write the Final Report.

5. Background Research

It will be necessary for you to do background research on your chosen project topic. Like every other aspect of the project the best way to achieve this is to produce a plan of what to do. In this case the best method is to produce a list of topics that you (and your Project Supervisor) think are relevant to your project. It is essential that you make notes on your background research. These notes will assist you when writing your Final Report.

6. How to Succeed

The key to success in the project unit is to write a work schedule, and then to adhere to it. You should prepare your work schedule in consultation with your project Supervisor, who will help you to make realistic estimates of how long various tasks will take. The project Supervisor's first job may be to cut down what you propose to a reasonable size. Do not be disheartened by this; when you are preparing a plan of work for the first time it is very easy to under-estimate how long things will take. It is better to produce a complete, well-rounded but medium-sized project than to make a botched job of something very ambitious. Remember it always takes longer than you think, therefore you should allow extra time in your schedule so that things like illness or unexpected machine failure do not totally ruin your schedule.

It is sometimes difficult to adhere to a plan of work because, as you acquire greater knowledge and expertise in the subject, you will probably think of other approaches to the problem, or extra features that you (or your client) would like to add in to your project. This is normal. If you really have to change some aspect of your project, discuss the matter with your project Supervisor first; keep records of your original approach and reasons for the change - these should be documented in the Final Report.

Throughout the course of the Project you are expected to adopt a professional attitude, taking responsibility for the management of the work according to the work schedule.

The next section outlines the contents of the project proposal, final report, presentation and demonstration.

7. The Project Proposal

The proposal is a blueprint for your project. The purpose of the proposal is to put forward your ideas about the project and allow your supervisor to judge whether the project is viable.

The project proposal should be between 4 and 6 pages in length and include the following sections:

7.1 Project background

An introduction providing context about your project: a brief discussion of the problem you are trying to solve; the source of the project and rationale; any research that informs the project and the scope of the project.

For example: 'The computing artefact to be developed is an Information System because … A small firm currently maintains its records in paper based form. This is inefficient for the following reasons ... The solution will reduce overheads such as ... It will potentially allow savings of £xxx per year and open up the following business opportunities ... The scope is limited to sales records because of time limitations.

7.2 The aims and objectives.

What is the overall aim of the project?

What are the specific objectives – stages to achieve the aim(s)? Are the objectives SMART?

For example: 'The aim is to develop a system that will automate the following transactions ... It will provide an online Help system supporting the following features ... Objectives are to: elicit detailed requirements, model the dbms as an ERD….

7.3 Method of solution / Architecture

How will you solve the problem/ achieve the aims? What technology, algorithms, methodology are appropriate (& why?)

E.g. Since the It will be developed in Java Version X and MySQL Version Y, to run on the following platforms ...'

7.4 Risk Assessment

Identify the main risks to your project, assess the likelihood and impact of each and rank by risk exposure. Show the results of the assessment in a table with an outline of how you will deal with them. Provide explanation of your prioritization.

7.5 Work Breakdown Structure and Gantt chart

This section needs to provide a breakdown of the main activities that you will need to undertake for the successful completion of your project. Indicate the key milestones with dates. *It is suggested that you use a tool such as Microsoft Project for producing this information.*

7.6 Project Deliverables (artefacts)

Describe the *project deliverables* (E.g. Software or hardware artefacts or reports).

The project will not proceed until the plan is approved by your supervisor.

8. The Project Report

It will take you longer than you think to write up your Final Project Report. You will also want to get your Supervisor to comment on a draft of the write-up; to allow sufficient time for the process of reading and revising the draft report. You should write sections of your report as you go along. This will need revision near the end, but it is far easier to revise work than write it from scratch.

Your project report is an account of the work done in terms of the development of the computing artefact. It is important that you plan for the report to communicate with the people reading it. This means producing a report that your supervisor will enjoy reading. The project report should be a well-structured word processed document that is easy to read. The project report should be 6000 (+/- 500) words in length.

It is important to point out that the content of your report will depend on the problem, method of solution and the tools used. Thus the models, implementation and testing could differ from those specified below. However, since computing projects are focussed on solving problems, the overall structure of the final report is unlikely to change. You should discuss the detailed content for your project with your tutor.

The structure of the project report is presented below, together with typical contents for a software development project.

8.1 Title Page

This section should contain the following centred information:

- The full title of the project
- The full name of the author, followed by the student registration number in brackets
- The centre at which the author is studying

8.2 Abstract

This section should consist of a synopsis of the project (150-200 words) stating the nature and scope of the work undertaken, method and a high level summary of the outcomes. It should enable a reader to decide whether to bother reading the rest of the report; no detail is required.

8.3 Contents Page

This section should show the page numbers of chapters, sections and sub-sections, a list of figures and tables, and a list of appendices.

8.4 Acknowledgements

This section is optional, but you may wish to pay tribute to particular people who have given you special assistance or support. However, there is little correlation between your final mark and the amount of praise lavished on your Supervisor!

8.5 Introduction

This chapter provides the context of your work in terms of:

- Background: discussion of the problem; the rationale; any research that informs the project and the scope of the project,
- The aims and objectives of the project. Objectives should be SMART,
- Method of solution with justification, overall architecture,
- Overview of the solution,
- A short overview of the remaining chapters.

Do not assume that the reader has specialised knowledge of the area that you are describing and be sure to explain any technical terms that you use.

Further guidance with respect to the content of this chapter is provided in Topic 5.

8.6 Analysis

This chapter consists of the analysis specification of the proposed system in terms of:

- Requirements Specification
    - A list of functional requirements
    - A list of system-wide non-functional requirements
    - The functional requirements should be prioritized using the MoSCoW prioritization technique.
    - A requirements catalogue should be included in an appendix.
- High level models for your system
    - Use Case
    - Architecture
    - Class diagrams
    - *Other models can be appropriate depending on your project*

Further guidance with respect to the content of this chapter is provided in Topic 2.

8.7 Design

This chapter consists of the design specification for the proposed system in terms of:

- Brief description of the purpose of the chapter and its contents, what the models show and why.
- Structural Model(s)
- Behavioural Model(s)
- User Interface Design
- Any frameworks used.
- Supporting text:
    - Explanation of each model; justification of choices
    - Explanation of choice of tools used for design and planned for implementation.

Further guidance with respect to the content of this chapter is provided in Topic 3.

8.8 Implementation

This chapter describes YOUR approach to implementing YOUR solution to the problem

- The contents can vary significantly depending on method / project type. E.g. for a software development project
    - Choice of programming language
    - System cutover from the development architecture to the implementation architecture
    - Data migration from the development architecture and/or existing systems to the implementation architecture
    - Training

Further guidance with respect to the content of this chapter is provided in Topic 5.

8.9 Testing

This chapter provides an account of your approach to testingfollowing the approach described in Topic 4. You need to describe what particular techniques you have used, why you have used these rather than others, and present a summary of your main results.

It should contain :

- Introduction: An overview of chapter contents and description of your approach to testing your system (Testing Strategy)
- An overview of how YOU developed YOUR first unit test – this should include the test script and an explanation of each field in the test script. Screen dumps should be provided to show how the first unit test was carried out.
- An overview of YOUR approach to integration and system testing
- Conclusion : A short evaluation of test results.

A Full Test plan should be incuded in the appendix

8.10 Conclusion

This chapter should begin with a summary of the aims of the project and how well you've achieved them and lessons learnt during the course of the project.

It must also contain the following sections:

Further Work: possible directions for further work should be given, and any outstanding problems should be described.

Critical Evaluation: this is a very important part of your Project Report. The aim of the critical evaluation is that you adopt a detached view of your Project and give an objective evaluation of your work. In particular you should look back and reflect on each of the main stages of the software development process:

Requirements: Were they complete, i.e. were all aspects of the system's behaviour described? Was anything missing? Were they too tightly linked to a particular design or implementation?

Design: Were alternatives considered at each stage? Were the correct alternatives chosen? Were the right sets of use cases, scenarios, classes, etc chosen?

Review of your plan and an explanation of any deviations from it.

Further guidance with respect to the content of this chapter is provided in Topic 5.

Reflective practices: Using the Rolfe, G., Freshwater, D. and Jasper, M. (2001) model, you should critically review the learning that you have undertaken in order to complete this project.

Based upon your learning, your reflection should include a description; an analysis and; an action plan in order to bring about improvements in the future.

NB. Appendix 3 provides useful reflection tasks for you to use whilst working on this project to ultimately prepare you for your written report.

8.11 References

All references you cite within the body of your report should be fully referenced in this section, using the Harvard Style. No reference should appear here unless it has been cited in the body of the report.

8.12 Appendices

Appendices may include any supporting material to which a reader might wish to refer, but which is not essential for the main body of the report. Appendices for the computing project may include some, or all, of the following:

- Requirements Catalogue
- Use Case Descriptions
- Detailed Class Definitions
- Test Scripts
- User Guide
- System Code
- External Client Documents where applicable (see Section 10)

9. The Project Presentation and Demonstration

Following completion of your project report, you will have 30 minutes to demonstrate the system developed and give a presentation about the content of the computing project. The demonstration and presentation will be assessed and contribute to your final mark.

The 30 minutes will be structured as follows:

- Demonstration (10 minutes)
- Questions related to the demonstration (5 minutes)
- Presentation (10 minutes)
- Questions related to the presentation (5 minutes)

The presentation should contain between 10 and 15 slides and should be structured as follows:

- An overview of the context/background of the project
- An overview of the analysis and design activities (high level model of the system, techniques used, key design decisions)
- An overview of the testing (techniques, a summary results, conclusions)
- A critique of the process (what went well, what didn't, what you have learned, what you would do differently next time)

PC and projection equipment will be available, so it is expected that you will give a PowerPoint presentation. The audience for the session will be your supervisor plus one other tutor.

10. Notes on Working with External Clients

You are not required to work with an external client to develop your project. However, if you are working with an external client the following must be included in an appendix:

- A letter of introduction from the Accredited Partner Centre to support the student.
- A memorandum of understanding (MOU) between the student and client. This document is typically prepared soon after a client has been chosen. Its purpose is to formally establish the specific details of the project work with the intention of protecting both the client and the student. It should cover:
  - A specification of the system project, including expected functionality, timeline, and resources to be committed.
  - Confidentiality of documents which may be passed from client to student.
  - Liabilities of student to client in case of failure to deliver a working product.
  - Financial compensation or remuneration. The MOU should make clear that the student is not to be paid for any project work undertaken.

Appendix 2: Project Assessment

There are four assessment components for the computing project:

- The Project Proposal: 10%
- The Project Report: 70%
- The Project Demonstration: 10%
- The Project Presentation: 10%

The Project Proposal: Grading criteria				
Assessment Criteria	1.1 Discuss project rationale, aims and objectives. 1.2 Select and justify an appropriate computing artefact to develop. 2.1 Select and justify the use of an appropriate development method. 2.2 Produce a project plan. 2.4 Identify and prioritise project risks.			
Mark	Project aims and objectives (25%)	Background/context / deliverable 25%)	Project plan (25%)	Risk Assessment (25%)
Fail (0-29)	Inadequate level of engagement with the task to demonstrate appropriate knowledge or skills.	Little or no discussion of rationale and project background.	No plan or irrelevant	No risk assessment or irrelevant.
Referral (30-39)	Proposal identifies some recognisably relevant aims and objectives.	Recognisable discussion of rationale and project background	Plan formatted and includes some relevant activities and attempt at Gantt chart.	Some recognisable risks identified showing relevance to project but major omissions/ misunderstanding.
Pass (40-59)	Aims and objectives & method identified in **outline** and are **broadly appropriate** for this level of study.	Discussion of rationale, project background & scope provides **some appropriate points**. Deliverables outlined.	Plan includes **some relevant activities**, milestones, Gantt chart but largely superficial & vague.	**Some relevant risks** identified with attempt at using impact and likelihood to prioritize.
Merit (60-69)	As pass and **most** aims objectives, method **are detailed**, **specific** and **achievable**.	As pass and **mostly detailed** and appropriate. **Detailed** deliverables.	As pass and plan is **detailed** and **almost complete** (may be some minor omissions / inaccuracies)	As pass and **impact and likelihood used correctly** to prioritize.
Distinction (70+)	As merit and **all** aims, objectives **specific**, consistently **detailed, achievable and unambiguous**. Method indicates how objectives or aims will be completed.	Discussion of rationale, project background & scope is **consistently detailed, convincing and unambiguous. Detailed deliverables.**	As merit and plan is **completely appropriate**.	As merit and correctly identifies **major risk factors and controls**

The Project Report: Grading criteria			
ILOs	2. Project manage the analysis, design, development and, where applicable, deployment of a computing artefact 3. Carry out the analysis for a computing artefact 4. Design and develop a computing artefact 5. Test a computing artefact 6. Systematically evaluate the activities undertaken to fulfil the aim of the project.		
	Project Report (80%)		Project Artefact (20%)
Mark	Process documentation (60%)	Evaluation/ Reflection (20%)	
Fail (0-29)	Inadequate level of engagement with the task to demonstrate appropriate knowledge or skills.	No , or irrelevant evaluation	No deliverable or not recognisably meeting what was planned
Referral (30-39)	Identifies some recognisably relevant content, including problem analysis and methods of solution.	Evaluation *describes* some problems and actions taken.	Deliverable(s) produced, showing some relevance to planned
Pass (40-59)	Report provides analysis of problem, context and describes methods of solution & testing with **some** recognised tools and techniques, though **level of detail is likely to be variable**. **Mostly** structured appropriately, omissions or inappropriate content in some sections. Some references resembling Harvard.	Evaluation describes some of: problems actions taken, learning achieved and what would be done differently, but with variable level of detail.	Deliverable(s) produced, meeting **some** of the requirements.
Merit (60-69)	As pass and **most** aspects **are appropriately detailed**. **All** sections structured appropriately containing **relevant content. All refs in Harvard**	A **balanced** and **detailed** reasoned evaluation of problems, actions, learning and alternative actions	As Pass and deliverables meet almost all requirements (maybe a few major weaknesses).
Distinction (70+)	As merit and **all** aspects are clear and appropriately detailed. Demonstrates a range of problem-solving approaches.	As merit and clear justification of conclusions.	As merit and deliverables meet all requirements (may be minor weaknesses)

The Project Demonstration: Grading criteria		
Assessment Criteria	1.1 Discuss project rationale, aims and objectives. 6.3 Articulate and demonstrate the outcomes of the project	
Mark	**Project meeting requirements (80%)**	**Demonstration planning and content (20%)**
Fail (0-29)	System is unusable or meets no requirements. User interface is inappropriate in several ways	Little or no evidence of planning, structure or alignment of demonstration with aims/objectives/requirements.
Referral (30-39)	System **partially meets some** requirements. User interface is **inconsistent but mostly appropriate**	Demonstration shows **recognisable alignment of demonstration with project aims**/ objectives requirements, but **major omissions** or unclear.
Pass (40-59)	System **fully meets some** functional requirements. User interface is **inconsistent but mostly appropriate**	Demonstration shows **planning** and aligns with **some** project aims/ objectives **requirements** but omissions or **lack of clarity**
Merit (60-69)	System **meets all requirements with minor weaknesses.** User interface is **consistent and recognisably employs good HCI principles**	Demonstration shows **clear and detailed planning** and **explicitly aligns** with **all project aims/ objectives** requirements but **minor omissions** or lack of clarity.
Distinction (70+)	System **exceeds functional requirements** . User interface is **consistent and recognisably employs good HCI principles**	Demonstration shows **clear and detailed planning,** with j**ustification of content** and **explicitly aligns** with all project aims/ objectives requirements.

The Project Presentation: Grading criteria		
Assessment Criteria	1.1 Discuss project rationale, aims and objectives. 6.1 Identify success criteria and judge strengths and weaknesses of the project process and outcomes. 6.2 Reflect on the process of conducting the project, identifying learning achieved 6.3 Articulate and demonstrate the outcomes of the project	
Mark	**Reflection/ evaluation (80%)**	**Process documentation discussion (20%)**
Fail (0-29)	No, or irrelevant evaluation & reflection.	Unclear or missing discussion of models, testing, background
Referral (30-39)	Reflection/ evaluation identifies some learning, though superficial. Question response shows limited engagement.	**Some recognisably relevant content,** including discussion of models, testing, background with **major omissions.**
Pass (40-59)	Reflection/ evaluation describes some of: strengths and weaknesses, learning achieved, but with variable level of detail. Some questions answered relevantly.	**Some** relevant content, including discussion of models, testing, background.
Merit (60-69)	Reflection/ evaluation provides focussed selection of: strengths and weaknesses, learning achieved, with appropriate detail Clear links to justify what would be done differently. Some minor weaknesses. Most questions answered succinctly, accurately, relevantly.	**All** relevant content, appropriate detail ,including **appropriate** discussion of models, testing, background. **Some minor weaknesses.**
Distinction (70+)	Reflection/ evaluation provides focussed selection of: **strengths and weaknesses, learning achieved**, with appropriate detail. Clear **links to justify** what would be done differently. **All** questions answered **succinctly, accurately, relevantly.**	**All** relevant content, appropriate detail, including **appropriate** discussion of models, testing, background.

Appendix 3: Reflective Practice

In the project report (Section 3.11), you are required to reflect upon the learning that has taken place whilst you have worked on this project. Prompts are given below in the form of questions to help focus your writing in this section. NB. You will not be expected to structure this by listing the questions and inserting the answers underneath, merely use this to guide your response.

The following identifies those stages in reviewing the progress of your computing project. Consider the those reflective questions at each stage to demonstrate your progress and learning on your computing project.

Planning Stage:

1. What are your motivations for undertaking the project topic in this area?
   Things to consider:
   - Why did you choose this topic and how is it relevant or why is it important to you?
   - Do you have a strong opinion or ideas on the topic you chosen?
   - Do you have a personal issue, problem or interest that you would like to know more about this topic?

2. Is the project topic you choose appropriate?

Things to consider:
- Is the topic manageable and material is available?
- Is the chosen topic feasible?
- Is the topic is relevance to the programme you study?

3. How might you go about undertaking this project topic?

Things to consider:
- How will you begin the research of the topic you are interested in?
- Do you have sufficient knowledge of the subject area to fulfil the criteria of project?
- Do you have access to appropriate sources for the topic chosen?

Background reading and analysis stage:

1. What have you learned about your project topic area so far?

Things to consider:
- Do you have sufficient data or information to analyse the current system?
- Is there sufficient focus to your project topic?
- Will your project topic allow you to think critically about the system you are researching and developing?

2. Given the data/information you have collected and analysed, do your need to reformulate your project direction?

Things to consider:
- What reasons may have affected your ability to continue the project as planned?
- To what extent does the data/information analysed relate to your proposed project topic?
- Are there any challenges that you need to overcome in order to achieve your desired outcomes?

Implementation and Design:

What do you like most about your proposed system?

- Is your system user friendly?
- Any special features that you are very proud off?
- Does your system reduce or solve any problems from the current system?

What do you like least about your proposed system?

- Any features you will like to add or enhanced to your proposed system?
- Is your system easy to use?
- Does your system can be used in any platforms?

What are the challenges you faced during system implementation? How you overcome them?
Things to consider:

- Did you struggle to complete any features of the system?
- Did you have enough time for coding?
- Did you ask for help when you were stuck?

Final stage: Conclusion

1. How do you feel your solution relates to real-world situations and problems?

Things to consider:

- What are the current situations and problems user facing?
- How your solution help to solve the problems faced?
- What features do you included in your solution for solving those issues?

2. What have you learned from this computing project unit? If you are going to do the project second time, anything you would do differently?

Things to consider:

- Did the computing project further your understanding of the programme you study?
- Did you check your project at different stages was correct before continuing?
- Have you analysed the mistakes your made during each stage of the project?
- Have you thought about any actions or steps that can improve your project?

Appendix 4: Project Assessment Forms

Assessment Form for Project Proposal (10%)

Student Name _____

Title of Project: _____

Supervisor: _____

The Project Proposal: Grading criteria (10%)					
Assessment Criteria		1.1 Discuss project rationale, aims and objectives. 1.2 Select and justify an appropriate computing artefact to develop. 2.1 Select and justify the use of an appropriate development method. 2.2 Produce a project plan. 2.4 Identify and prioritise project risks.			
	Mark Range	Project aims and objectives (25%)	Background/context / deliverable 25%)	Project plan (25%)	Risk Assessment (25%)
Fail (0-29)	0-7	Inadequate level of engagement with the task to demonstrate appropriate knowledge or skills.	Little or no discussion of rationale and project background.	No plan or irrelevant	No risk assessment or irrelevant.
Referral (30-39)	8 -9	Proposal identifies some recognisably relevant aims and objectives.	Recognisable discussion of rationale and project background	Plan formatted and includes some relevant activities and attempt at Gantt chart.	Some recognisable risks identified showing relevance to project but major omissions/ misunderstanding.
Pass (40-59)	10 - 14	Aims and objectives & method identified in **outline** and are **broadly appropriate** for this level of study.	Discussion of rationale, project background & scope provides **some appropriate points**. Deliverables outlined.	Plan includes **some relevant activities**, milestones, Gantt chart but largely superficial & vague.	**Some relevant risks** identified with attempt at using impact and likelihood to prioritise.
Merit (60-69)	**15 - 17**	As pass and **most** aims objectives, method **are detailed**, **specific** and **achievable**.	As pass and **mostly detailed** and appropriate. **Detailed** deliverables.	As pass and plan is **detailed** and **almost complete** (may be some minor omissions / inaccuracies)	As pass and **impact and likelihood used correctly** to prioritize.
Distinction (70+)	18 - 25	As merit and **all** aims, objectives **specific**, consistently **detailed, achievable and unambiguous**. Method indicates how objectives or aims will be completed.	Discussion of rationale, project background & scope is **consistently detailed, convincing and unambiguous. Detailed deliverables.**	As merit and plan is **completely appropriate**.	As merit and correctly identifies **major risk factors and controls**
Marks					
Total Marks					

Supervisor comments:

Signature of supervisor: _____ Date: _____

Assessment Form for Project Report & Final Artefact (70%)

Student Name: _____

Title of Project: _____

Supervisor: _____

Project Report & Final Artefact (70%)

The Project Report & Final Artefact Grading criteria						
ILOs	2. Project manage the analysis, design, development and where applicable, deployment of a computing artefact 3. Carry out the analysis for a computing artefact 4. Design and develop a computing artefact 5. Test a computing artefact 6. Systematically evaluate the activities undertaken to fulfil the aim of the project.					
	Project Report (80%)				Project Artefact (20%)	
Mark	Process documentation (60%)	Mark Range	Evaluation/ Reflection (20%)	Mark Range		
Fail (0-29)	Inadequate level of engagement with the task to demonstrate appropriate knowledge or skills.	0 - 17	No, or irrelevant evaluation	0-5	No deliverable or not recognisably meeting what was planned	0-5
Referral (30-39)	Identifies some recognisably relevant content, including problem analysis and methods of solution.	18-23	Evaluation *describes* some problems and actions taken.	6-7	Deliverable(s) produced, showing some relevance to planned	6-7
Pass (40-59)	Report provides analysis of problem, context and describes methods of solution & testing with **some** recognised tools and techniques, though **level of detail is likely to be variable**. **Mostly** structured appropriately, omissions or inappropriate content in some sections. **Most references in Harvard.**	24-35	Evaluation describes some of: problems actions taken, learning achieved and what would be done differently, but with variable level of detail.	8-11	Deliverable(s) produced, meeting **some** of the requirements.	8-11
Merit (60-69)	As pass and **most** aspects **are appropriately detailed**. **All** sections structured appropriately containing **relevant content. All refs in Harvard**	36-41	A **balanced** and **detailed** reasoned evaluation of problems, actions, learning and alternative actions	12-13	As Pass and deliverables meet almost all requirements (maybe a few major weaknesses).	12-13
Distinction (70+)	As merit and **all** aspects are clear and appropriately detailed. Demonstrates a range of problem-solving approaches.	42-60	As merit and clear justification of conclusions.	14-20	As merit and deliverables meet all requirements (may be minor weaknesses)	14-20
	Marks					
	Total					

Supervisor comments:

Signature of supervisor: _____ Date: _____

Project Demonstration (10%)

Student Name: _____

Title of Project: _____

Supervisor: _____

The Project Demonstration: Grading criteria (10%)				
Assessment Criteria	1.1 Discuss project rationale, aims and objectives. 6.3 Articulate and demonstrate the outcomes of the project			
Mark	**Project meeting requirements (80%)**	**Mark Range**	**Demonstration planning and content (20%)**	**Mark Range**
Fail (0-29)	System is unusable or meets no requirements. User interface is inappropriate in several ways	0 - 23	Little or no evidence of planning, structure or alignment of demonstration with aims/objectives/requirements.	0-5
Referral (30-39)	System **partially meets some** requirements. User interface is **inconsistent but mostly appropriate**	24 -31	Demonstration shows **recognisable alignment of demonstration with project aims**/ objectives requirements, but **major omissions** or unclear.	6-7
Pass (40-59)	System **fully meets some** functional requirements. User interface is **inconsistent but mostly appropriate**	32-47	Demonstration shows **planning** and aligns with **some** project aims/ objectives **requirements** but omissions or **lack of clarity**	8-11
Merit (60-69)	System **meets all requirements with minor weaknesses.** User interface is **consistent and recognisably employs good HCI principles**	48-55	Demonstration shows **clear and detailed planning** and **explicitly aligns** with **all project aims/ objectives** requirements but **minor omissions** or lack of clarity.	12-13
Distinction (70+)	System **exceeds functional requirements** . User interface is **consistent and recognisably employs good HCI principles**	56-80	Demonstration shows **clear and detailed planning,** with j**ustification of content** and **explicitly aligns** with all project aims/ objectives requirements.	14-20
	Marks			
	Total Marks			

Supervisor comments:

Signature of supervisor: _____ Date: _____

Project Presentation (10%)

Student Name: _____

Title of Project: _____

Supervisor: _____

The Project Presentation: Grading criteria				
Assessment Criteria	1.1 Discuss project rationale, aims and objectives. 6.1 Identify success criteria and judge strengths and weaknesses of the project process and outcomes. 6.2 Reflect on the process of conducting the project, identifying learning achieved 6.3 Articulate and demonstrate the outcomes of the project			
Mark	**Reflection/ evaluation (80%)**	**Mark Range**	**Process documentation discussion (20%)**	**Mark Range**
Fail (0-29)	No, or irrelevant evaluation & reflection.	0 - 23	Unclear or missing discussion of models, testing, background	0-5
Referral (30-39)	Reflection/ evaluation identifies some learning, though superficial. Question response shows limited engagement.	24 -31	**Some recognisably relevant content,** including discussion of models, testing, background with **major omissions.**	6-7
Pass (40-59)	Reflection/ evaluation describes some of: strengths and weaknesses, learning achieved, but with variable level of detail. Some questions answered relevantly.	32-47	**Some** relevant content, including discussion of models, testing, background.	8-11
Merit (60-69)	Reflection/ evaluation provides focussed selection of: strengths and weaknesses, learning achieved, with appropriate detail Clear links to justify what would be done differently. Some minor weaknesses. Most questions answered succinctly, accurately, relevantly.	48-55	**All** relevant content, appropriate detail ,including **appropriate** discussion of models, testing, background. **Some minor weaknesses.**	12-13
Distinction (70+)	Reflection/ evaluation provides focussed selection of: **strengths and weaknesses, learning achieved**, with appropriate detail. Clear **links to justify** what would be done differently. **All** questions answered **succinctly, accurately, relevantly.**	56-80	**All** relevant content, appropriate detail, including **appropriate** discussion of models, testing, background.	14-20
	Marks			
	Total Marks			

Supervisor comments:

Signature of supervisor: _____ Date: _____